

SYNTHETIC DATA PREPARED FOR AGRONET

Generic Fertilizer Handbook for Indian Agriculture

This handbook is intended for educational use and RAG testing. It summarizes commonly accepted fertilizer management practices in India based on agricultural extension guidance and broadly applicable recommendations. Actual fertilizer application should always follow local soil-test results and official advisories.

1. Introduction

Fertilizers supply essential nutrients required for healthy crop growth. Balanced nutrient management improves yield, crop quality, soil fertility, and long-term sustainability.

2. Essential Plant Nutrients

Macronutrients:

- Nitrogen (N): Promotes leafy growth and chlorophyll formation.
- Phosphorus (P): Encourages root development, flowering and early growth.
- Potassium (K): Improves plant vigor, water regulation and stress tolerance.

Secondary nutrients:

- Calcium (Ca)
- Magnesium (Mg)
- Sulphur (S)

Micronutrients:

- Zinc, Iron, Boron, Copper, Manganese, Molybdenum and Chlorine are required in smaller quantities.

3. Common Fertilizers Used in India

- Urea – high nitrogen fertilizer.
- DAP (Diammonium Phosphate) – nitrogen and phosphorus.
- MOP (Muriate of Potash) – potassium source.
- SSP (Single Super Phosphate) – phosphorus with sulphur.
- NPK Complex Fertilizers – balanced nutrient formulations.
- Organic manures – Farmyard manure, compost, vermicompost.
- Biofertilizers – Rhizobium, Azotobacter, Azospirillum, PSB and mycorrhizae.

4. Soil Testing

A soil test should be performed before deciding fertilizer doses.

Benefits:

- Identifies nutrient deficiencies

- Prevents excessive fertilizer use
- Improves fertilizer efficiency
- Saves production costs
- Supports sustainable agriculture

5. Fertilizer Application Methods

- Basal application before sowing
- Band placement near seed rows
- Top dressing during crop growth
- Foliar spray for selected nutrients
- Fertigation through drip irrigation systems

6. Integrated Nutrient Management (INM)

INM combines chemical fertilizers with organic manures, crop residues and biofertilizers.

Advantages:

- Better soil health
- Higher nutrient use efficiency
- Reduced environmental impact
- Improved microbial activity
- Sustainable productivity

7. Common Nutrient Deficiency Symptoms

Nitrogen: pale green leaves, reduced growth.

Phosphorus: stunted plants, dark green or purplish foliage.

Potassium: yellowing and scorching of leaf margins.

Zinc: shortened internodes and chlorosis.

Iron: yellowing of young leaves while veins remain green.

Boron: poor flowering and fruit development.

8. Safe Fertilizer Handling

- Store fertilizers in a cool, dry place.
- Keep bags away from moisture.
- Use personal protective equipment when handling dusty materials.
- Do not mix incompatible fertilizers unless recommended.
- Keep fertilizers away from children, livestock and drinking water sources.

9. Government Initiatives

Important initiatives include:

- Soil Health Card Scheme
- Neem-coated Urea
- Nutrient-Based Subsidy (NBS)

- Promotion of balanced fertilization
- Integrated Nutrient Management through agricultural extension programs

10. Best Practices

- Follow soil-test recommendations.
- Apply fertilizers at the correct time.
- Split nitrogen applications where appropriate.
- Incorporate organic matter regularly.
- Avoid over-application.
- Maintain proper irrigation after fertilizer application when required.
- Keep records of fertilizer use each season.

11. Summary

Balanced fertilizer management improves productivity while protecting soil health. Combining soil testing, efficient nutrient application, organic amendments and government-recommended best practices helps maintain sustainable agricultural production.